LIVE

Diffraction

" is bending of a wave when it encounters a small opening or obstacle



Hello Sir

RI Rishi Dalal
ncert

SA Sanyam kumar Jain
ncert

RI Rishi Dalal
apke mardarshan

AR Arnab Goyal
good morning sir

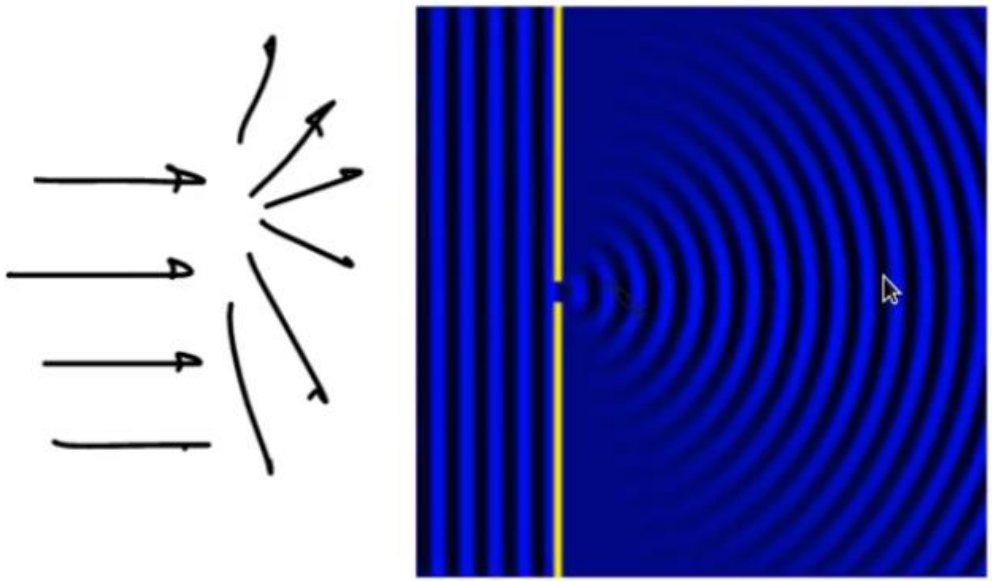
DE Deepak Chaurasia
Chem padh ke physics likh diye

PR Pranesh K.r
sir yes

Type a message

Chat Doubts More

LIVE





RI Rishi Dalal
ncert

SA Sanyam kumar Jain
ncert

RI Rishi Dalal
apke mardarshan

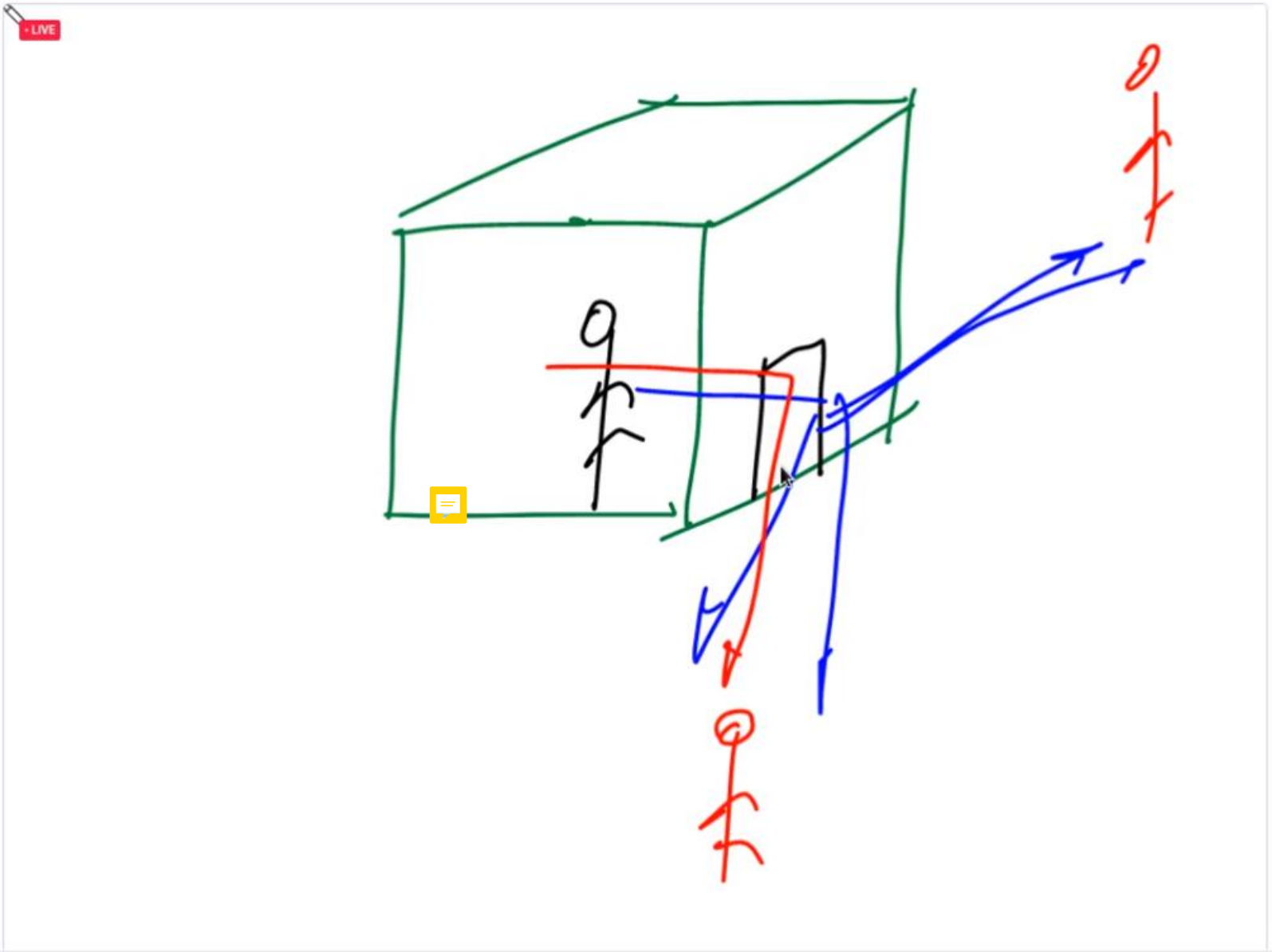
AR Arnab Goyal
good morning sir

DE Deepak Chaurasia
Chem padh ke physics likh diye

PR Pranesh K.r
sir yes

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RI Rishi Dalal
apke mardarshan

AR Arnab Goyal
good morning sir

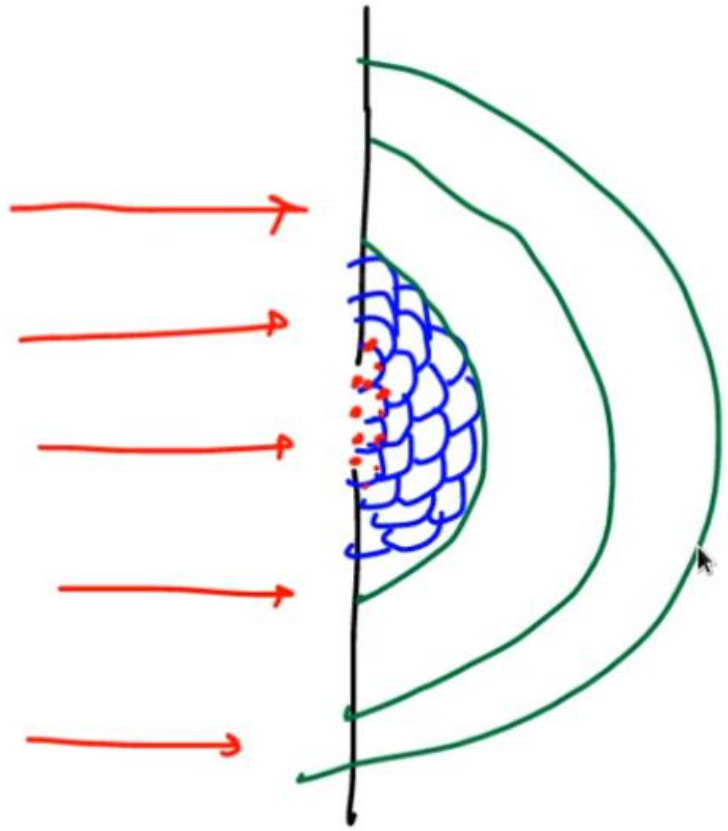
DE Deepak Chaurasia
Chem padh ke physics likh diye

PR Pranesh K.r
sir yes

SH Shardul Date
oh the red people arent able to see inside the door

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Chat Doubts More





Haan

RI Rishi Dalal
yes

RA Rachit Jain
Yes sir

SA Sanyam kumar Jain
y

SA Sashi Raam v v
y

AM Amna Rahman
yes

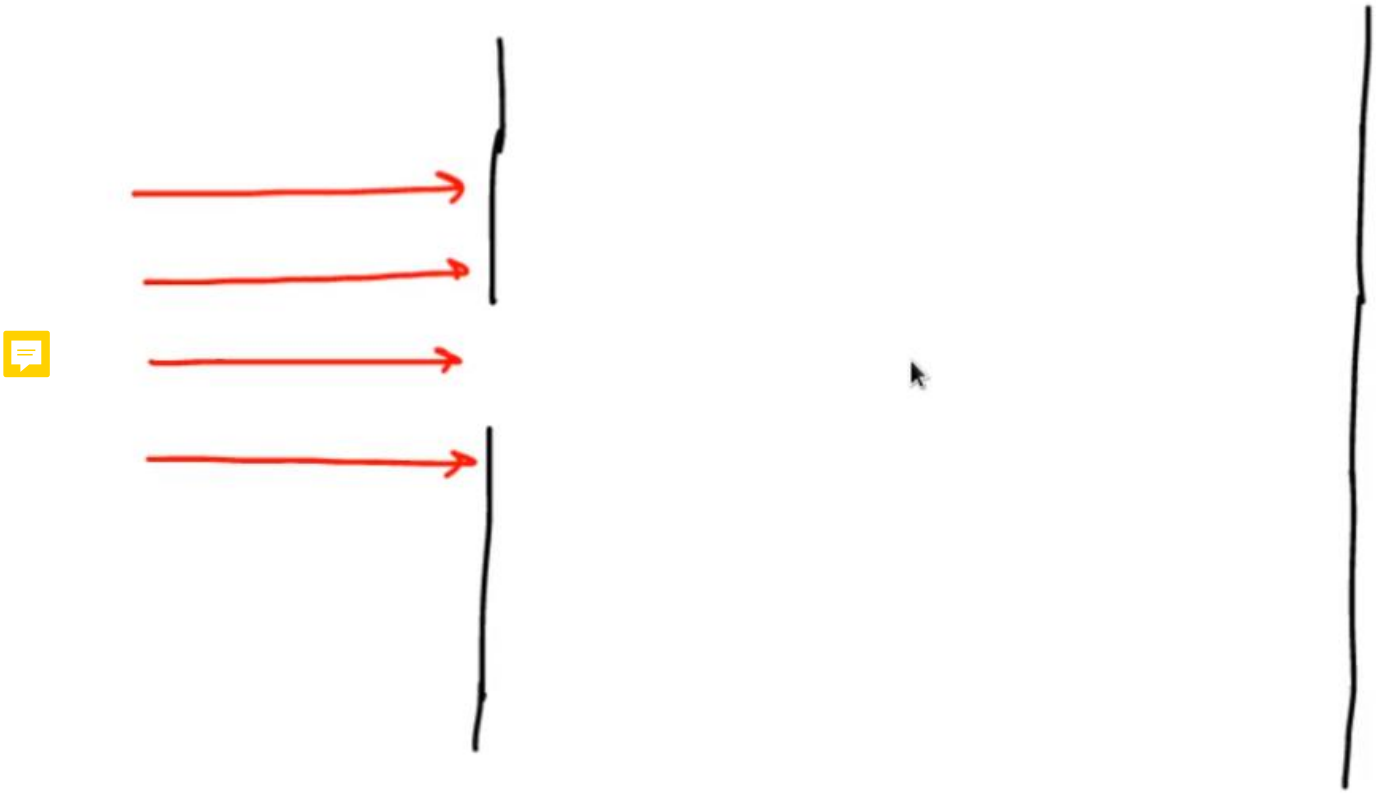
SR Srijahnavi Ch
y

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Chat Doubts More

LIVE

Fraunhofer diffraction / Single slit diffraction





AM Amna Rahman
yes

SR Srijahnavi Ch
y

RI Rishav Raj
cyl

RA Rachit Jain
cylinder

AB Abhiharsh Chauhan
far field diffration too

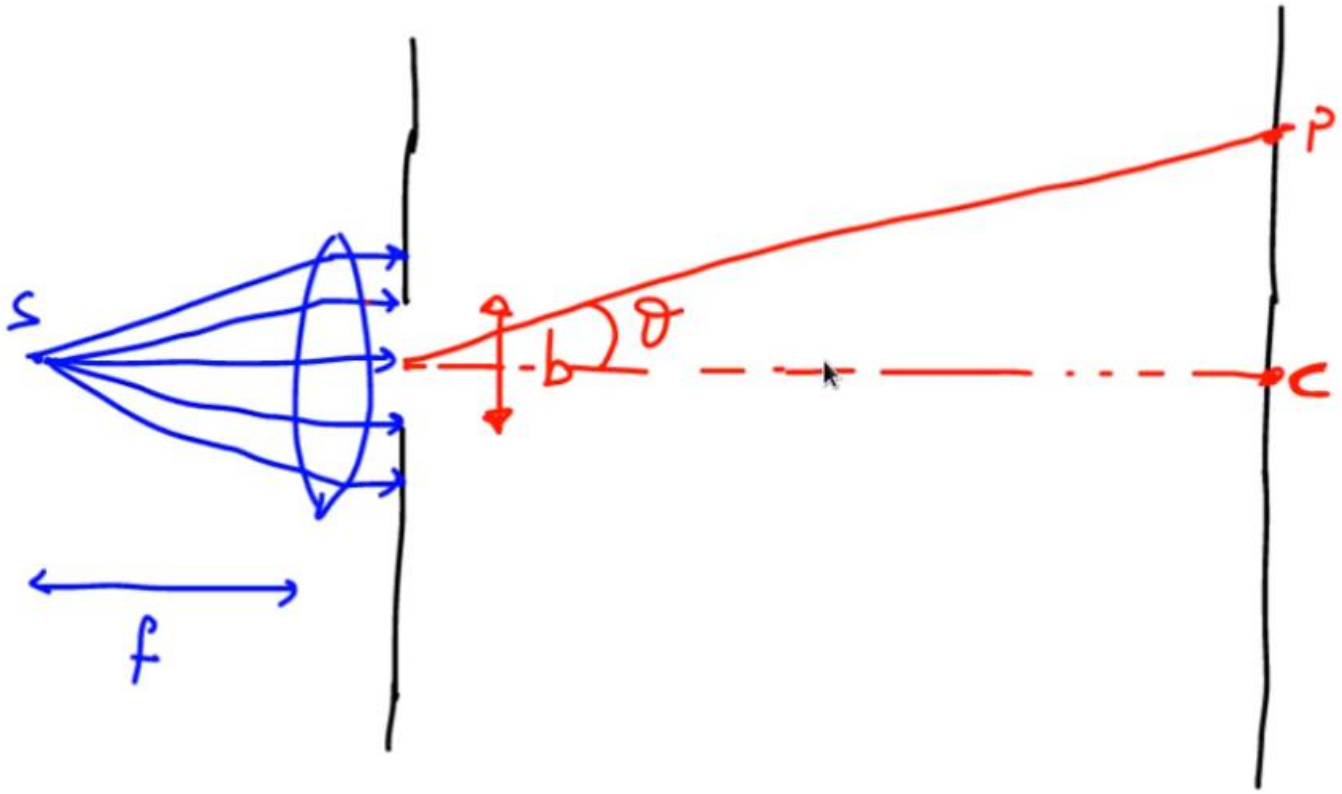
RI Rishi Dalal
source is at inf dist

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Chat Doubts More

LIVE

Fraunhofer diffraction / Single slit diffraction





Rishav Raj
cyl

Rachit Jain
cylinder

Abhiharsh Chauhan
far field diffration too

Rishi Dalal
source is at inf dist

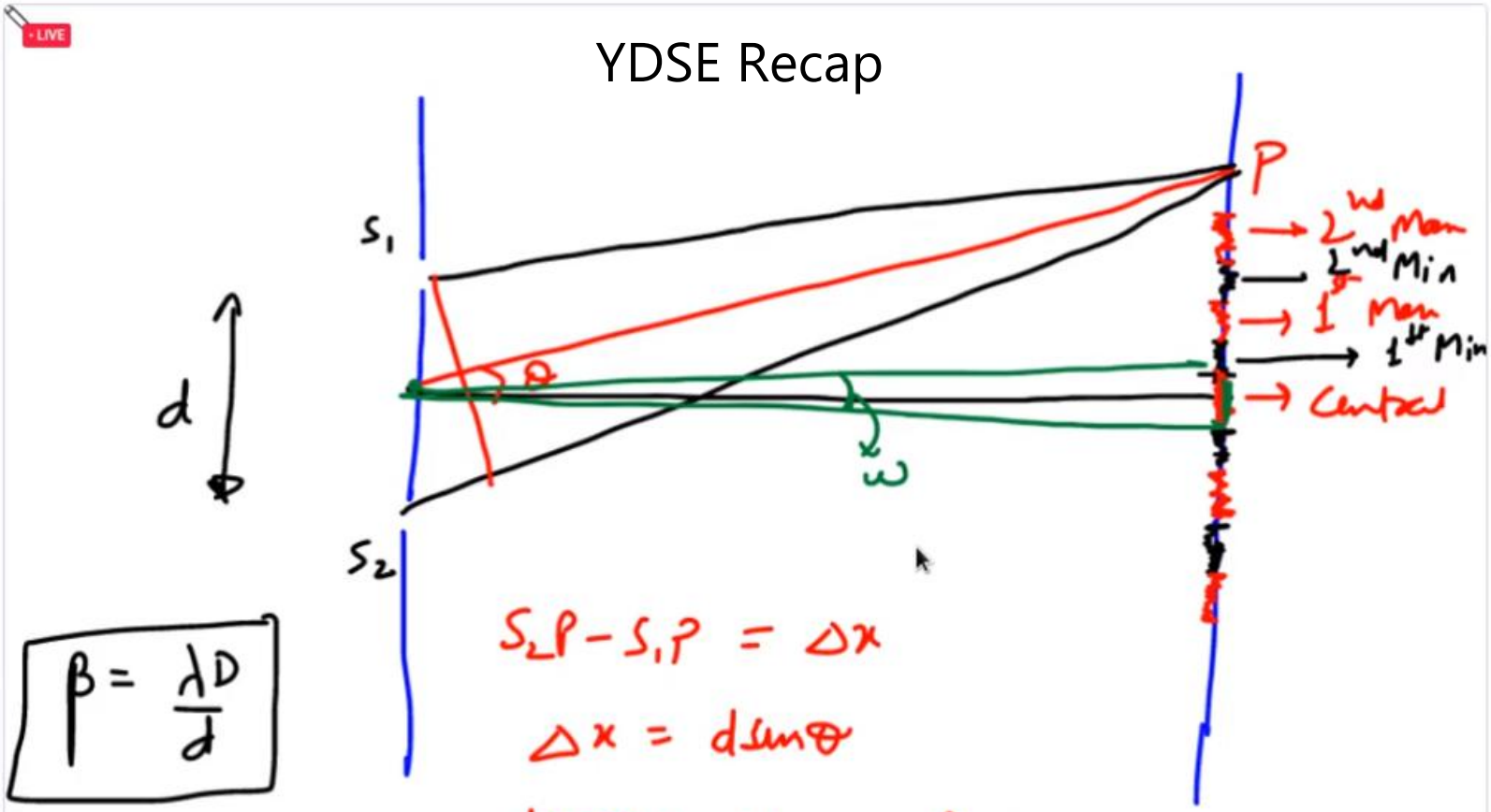
Rishi Dalal
convex lens

Rishi Dalal
f

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Chat Doubts More

YDSE Recap



$$\beta = \frac{\lambda D}{d}$$

$$\omega = \frac{\beta}{D} = \frac{\lambda}{d}$$

$$S_2P - S_1P = \Delta x$$

$$\Delta x = d\sin\theta$$

$$d\sin\theta = n\lambda \rightarrow \text{for max}$$

$$d\sin\theta = \left(n - \tfrac{1}{2}\right)\lambda \rightarrow \text{for min.}$$



Jashan Bhatia Insan
y

RI Rishi Dalal
nl

RA Rachit Jain
Y

DE Deepak Chaurasia
Y

AB Abhiharsh Chauhan
beta/D

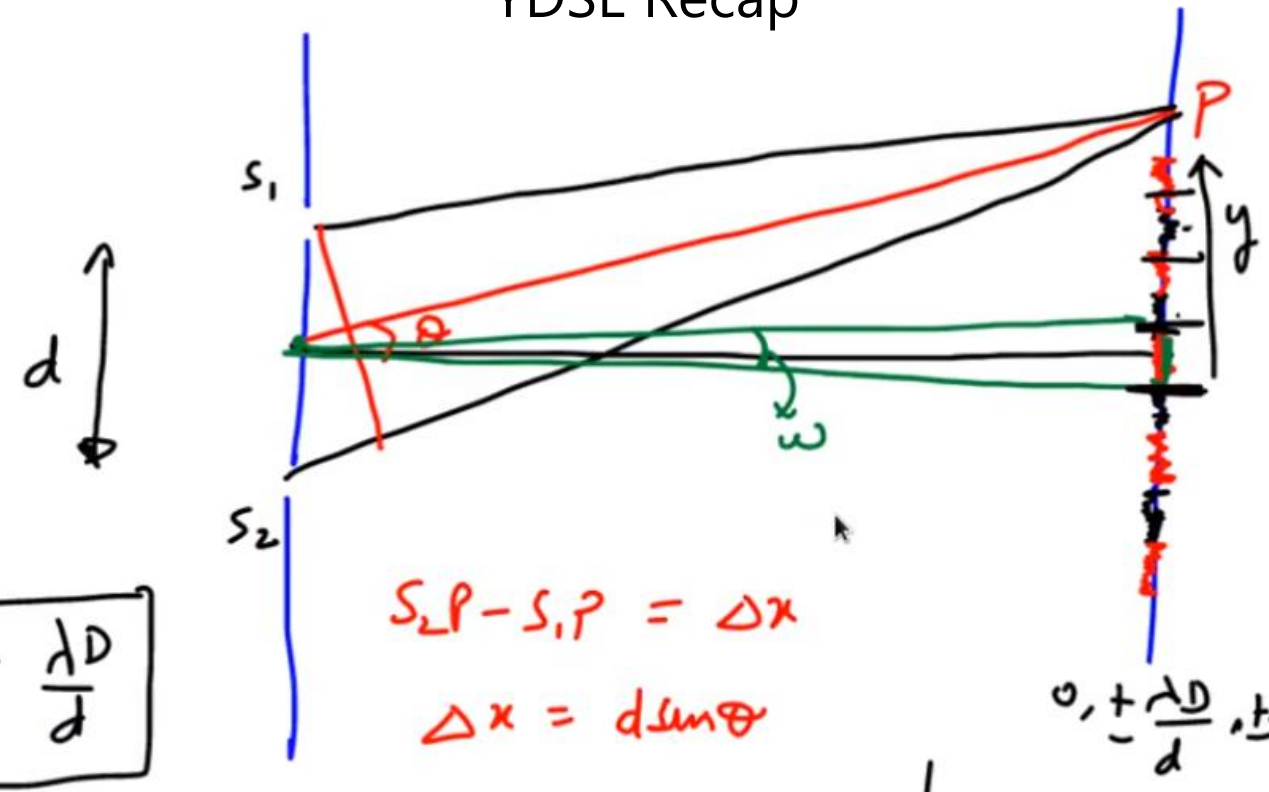
RI Rishi Dalal
yes

RA Rachit Jain
Y

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Chat Doubts More

YDSE Recap



$$\beta = \frac{\lambda D}{d}$$

$$S_2P - S_1P = \Delta x$$

$$\Delta x = d\sin\theta$$

$$0, \pm\frac{\lambda D}{d}, \pm 2\frac{\lambda D}{d}$$

$$\omega = \frac{\beta}{D} = \frac{\lambda}{d}$$

$$\frac{y}{D} = \sin\theta$$

$$\Delta x = \frac{dy}{D}$$

$$n\lambda = \frac{dy}{D}$$

$$y = \frac{n\lambda D}{d}$$



DE Deepak Chaurasia
Y

AB Abhiharsh Chauhan
beta/D

RI Rishi Dalal
yes

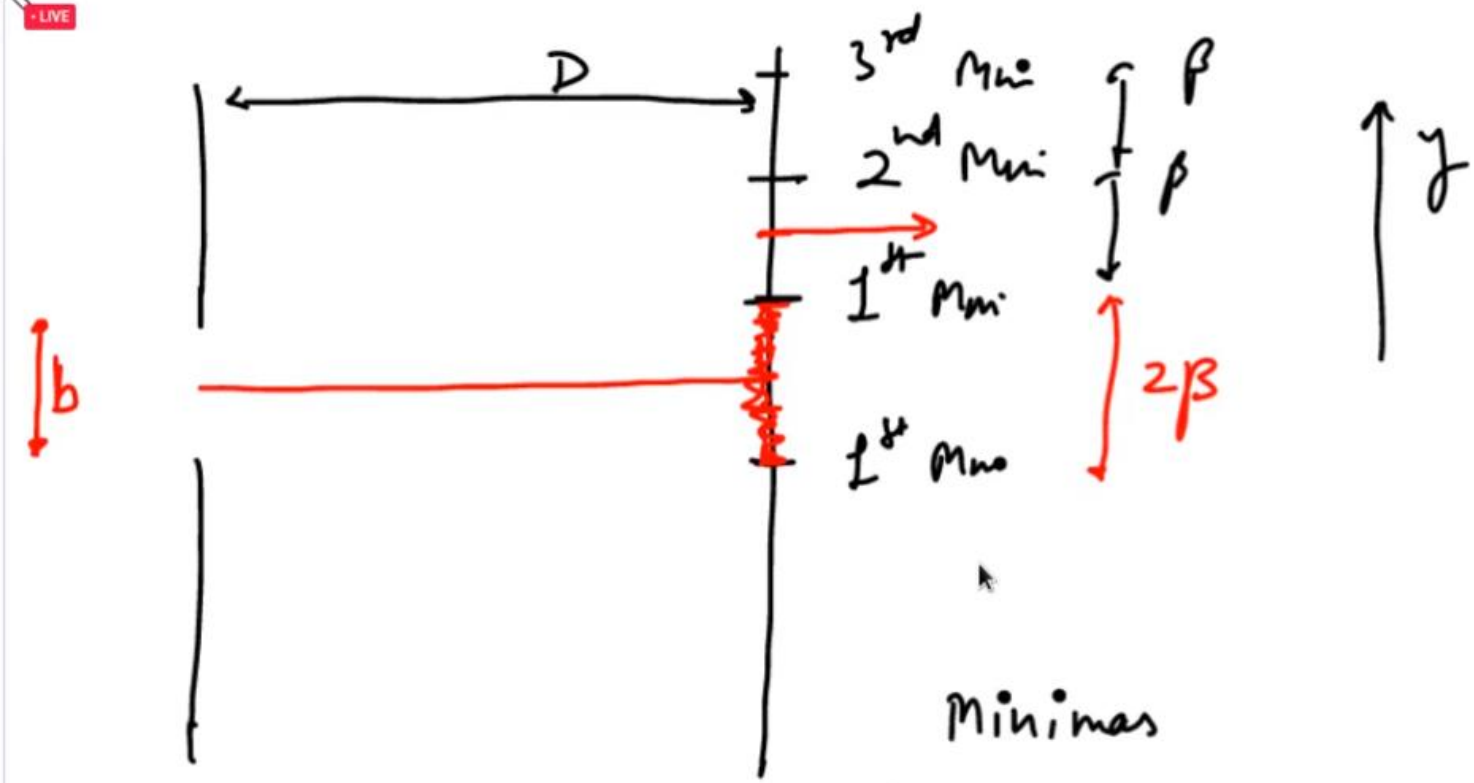
RA Rachit Jain
Y

RI Rishi Dalal
2

SR Srijahnavi Ch
btea is also wdith of central maxima ryt?

Type a message

Chat Doubts More



Minimas

$$y = \pm\frac{\lambda D}{b}, \pm\frac{2\lambda D}{b}, \pm\frac{3\lambda D}{b}$$

$$2\beta = \frac{2\lambda D}{b}$$ = Width of Central Maxima = Width of diffraction pattern

Maximas

$$y = \pm 1.5\frac{\lambda D}{d}, \pm 2.5\frac{\lambda D}{d} \ldots$$



Rachit Jain: y

Amna Rahman: yes

Sashi Raam v v: y

Jashan Grover insan: y

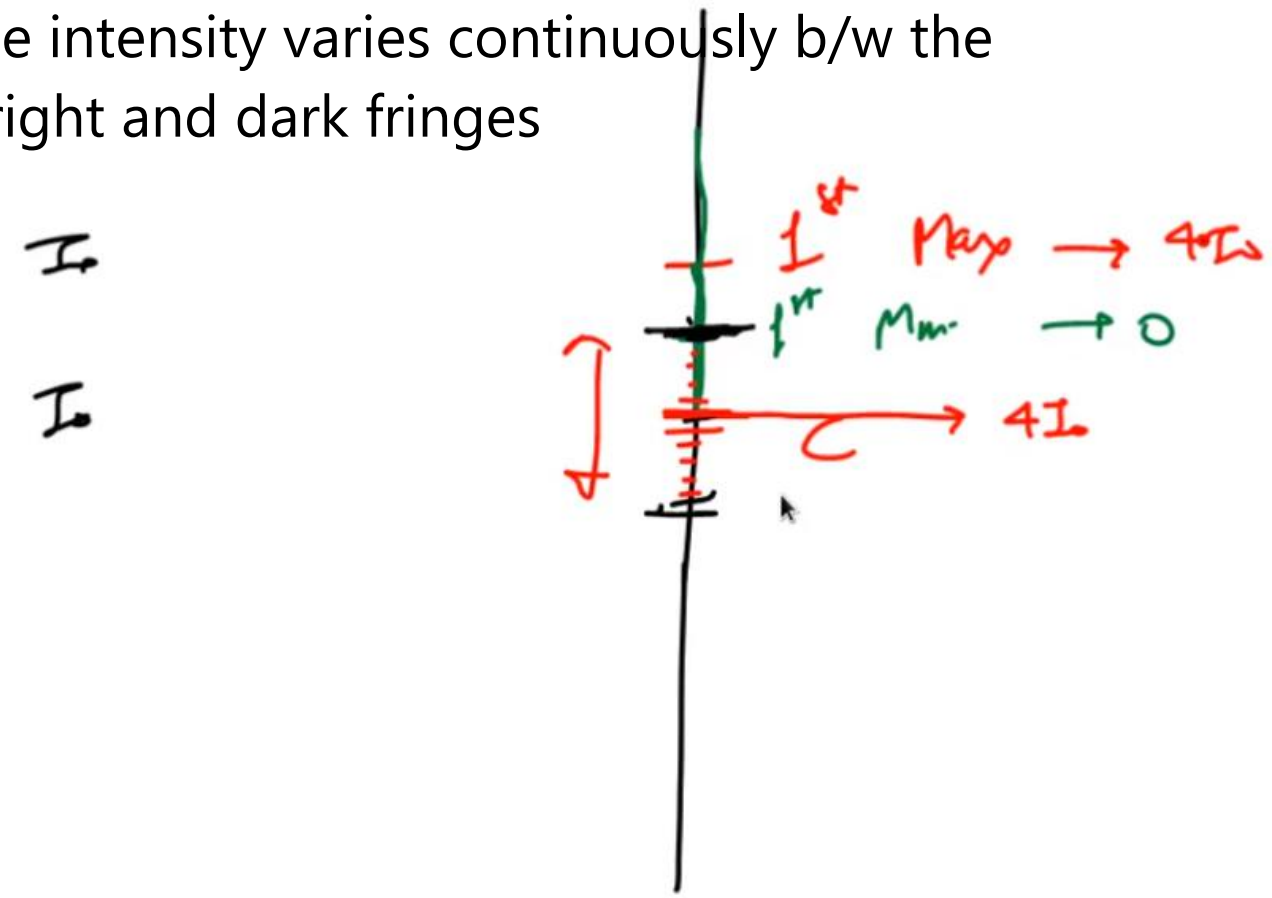
Pranesh K.r: yes sir

Rishi Dalal: sir so 1.5 aur 2 ke beech mein kya hoga?

Type a message

Chat Doubts More

the intensity varies continuously b/w the bright and dark fringes





ke beech mein kya hoga?

RI Rishav Raj
varying hogi

RI Rishi Dalal
haa

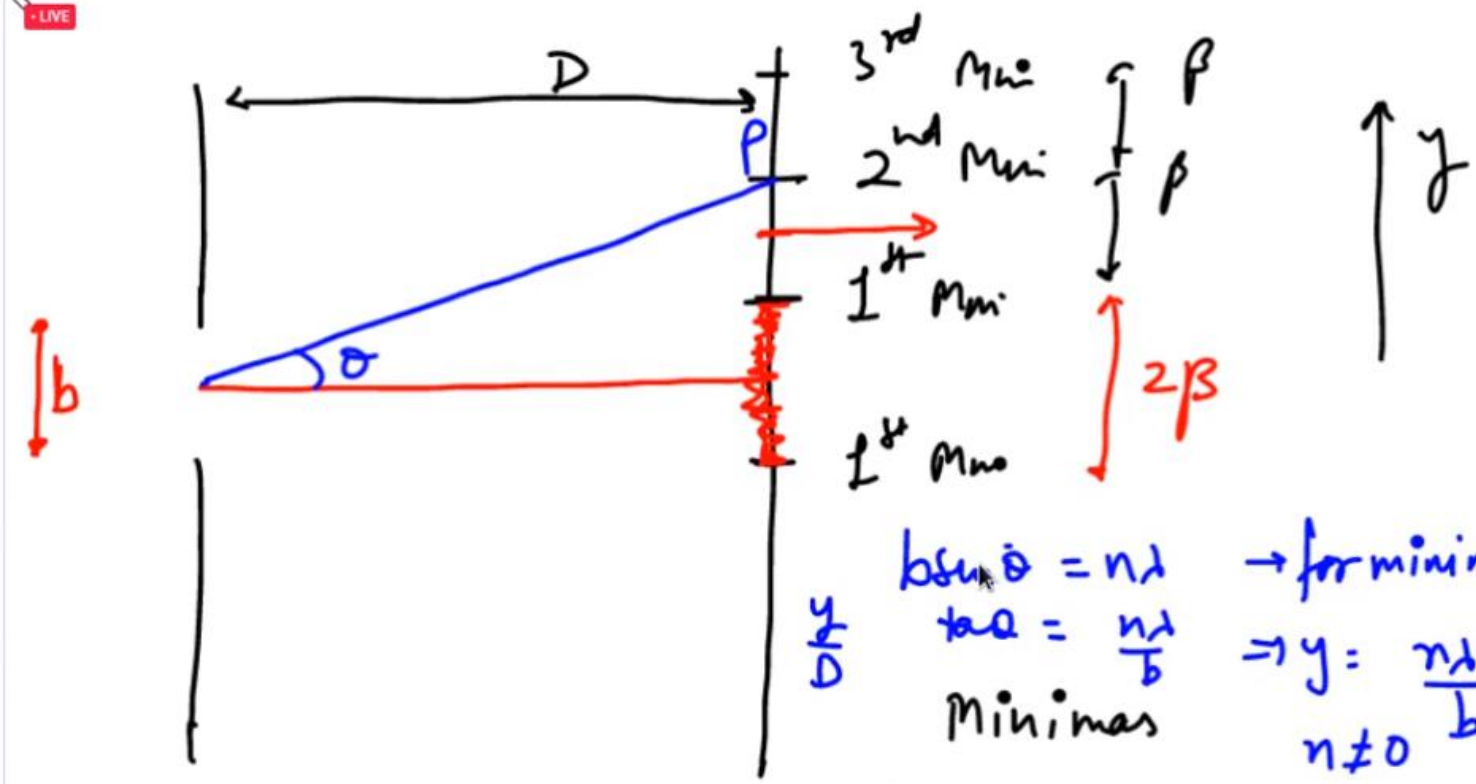
RI Rishi Dalal
ab aa gaya

JA Jashan Grover insan
central point is considerd as max min

RI Rishi Dalal
yes

Type a message

Chat Doubts More



$b\sin\theta = n\lambda$ → for minima

$\frac{y}{D}$ $\tan\theta = \frac{n\lambda}{b}$ $\Rightarrow y = \frac{n\lambda D}{b}$

Minimas $n \neq 0$

$y = \pm\frac{\lambda D}{b}, \pm\frac{2\lambda D}{b}, \pm\frac{3\lambda D}{b}$

$2\beta = \frac{2\lambda D}{b}$ = Width of Central Maxima

$\omega = \frac{2\beta}{D} = \frac{2\lambda}{b}$ = Width of diffraction pattern

Maximas

$y = \pm 1.5\frac{\lambda D}{d}, \pm 2.5\frac{\lambda D}{d} \ldots$



yes

AM Amna Rahman: yes

DE Deepak Chaurasia: Y

RI Rishi Dalal: L/b

AM Amna Rahman: 2beta/D

HA Harshal Sethiya: 2 lambda/b

RI Rishi Dalal: haa

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Chat Doubts More

LIVE

A single slit of width b is illuminated by a coherent monochromatic light of wavelength λ. If the second and fourth minima in the diffraction pattern at a distance 1m from the slit are at 3cm and 6cm respectively from the central maximum, what is the width of the central maximum? (i.e. distance between first minimum on either side of the central maximum)

|2017|

(a) 1.5 cm
(b) 3.0 cm
(c) 4.5 cm
(d) 6.0 cm

$4\beta - 2\beta = 6 - 3$

$= 3\text{cm}$

$2\beta = 3\text{cm}$

$\beta = 1.5\text{cm} = \frac{\lambda D}{b}$



HA Harshaditya
2 lambda/b

RI Rishi Dalal
haa

RI Rishi Dalal
b

SA Sanyam kumar Jain
B

RA Rachit Jain
b

AB Abhiharsh Chauhan
b

MA Manoj
B

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Chat Doubts More

LIVE

b

A single slit of width 0.1 mm is illuminated by a parallel beam of light of wavelength 6000 Å and diffraction bands are observed on a screen 0.5m from the slit. The distance of the third dark band from the central bright band is: |2017|

D

(a) 3 mm (b) 9 mm

(c) 4.5 mm (d) 1.5 mm

$\frac{3\lambda D}{b}$



Srijahnavi Ch
yes

MA Madhav
y

SA Sanyam kumar Jain
Y

MA Manoj
Yes

HA Harshal Sethiya
Y

Jatin Gupta

MA Madhav Lata
B

Type a message

Chat Doubts More

LIVE

The box of a pin hole camera, of length L, has a hole of radius a. It is assumed that when the hole is illuminated by a parallel beam of light of wavelength λ the spread of the spot (obtained on the opposite wall of the camera) is the sum of its geometrical spread and the spread due to diffraction. The spot would then have its minimum size (say b_{min}) when: [2016]

$x = \frac{2\lambda D}{2a} + 2a$

$x = \frac{\lambda D}{a} + 2a$

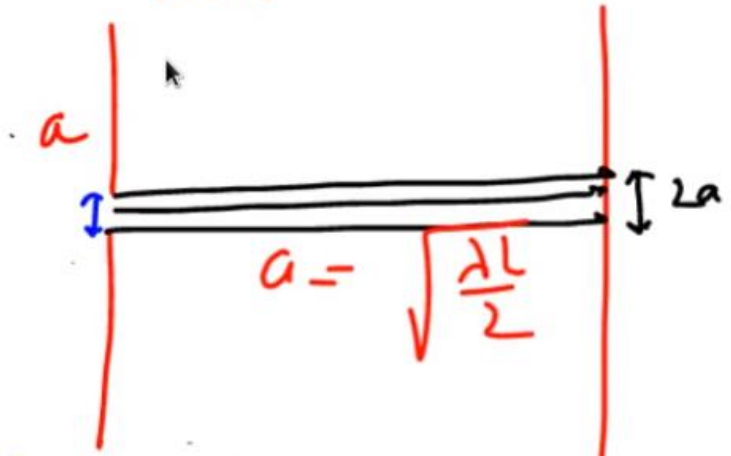
radius

(a) $a = \sqrt{\lambda L}$ and $b_{min} = \left(\frac{2\lambda^2}{L}\right)$

(b) $a = \sqrt{\lambda L}$ and $b_{min} = \sqrt{4\lambda L}$

(c) $a = \frac{\lambda^2}{L}$ and $b_{min} = \sqrt{4\lambda L}$

(d) $a = \frac{\lambda^2}{L}$ and $b_{min} = \left(\frac{2\lambda^2}{L}\right)$



$\frac{2\lambda L}{2a} + 2a = x$

$x = 2a + \frac{\lambda L}{a}$



radius) ki value niklegi

AM Amna Rahman: it is width of central maxima @ srijahnavi

JA Jashan Grover insan: harshal still it will be same

SR Srijahnavi Ch: ok thanks

RI Rishav Raj: sir ans shaayad radius i.e x/2 se nikaala hai

Type a message

Chat Doubts More

LIVE

The angular width of the central maximum in a single slit diffraction pattern is 60°. The width of the slits is 1 μm. The slit is illuminated by monochromatic plane waves. If another slit of same width is made near it, Young's fringes can be observed on a screen placed at a distance 50 cm from the slits. If the observed fringe width is 1 cm, what is slit separation distance? [2018]

(a) 75 μm (b) 100 μm

(c) 25 μm (d) 50 μm

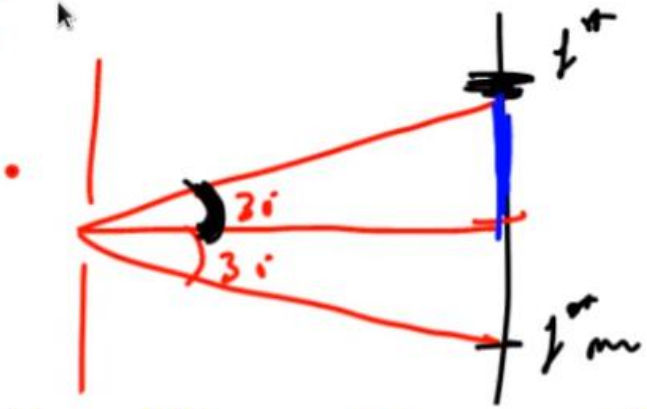
$\beta = \frac{\lambda D}{d} = 1\,cm$

$10^{-6} \times \frac{\pi}{6} \times \frac{1}{2} \times \frac{1}{d} = 1\,cm$

$b \sin\theta = \lambda$

$1\,\mu m \times \frac{1}{2} = \lambda$

$\lambda = \frac{1}{2}\,\mu m$



$\beta = \frac{\lambda D}{d} \Rightarrow 10^{-2} = \frac{1}{2} \times \frac{50 \times 10^{-2}}{d}$

$d = 25$



sir bsintheta = lamda lo as angle is large here

MA Madhav: pi =3.14

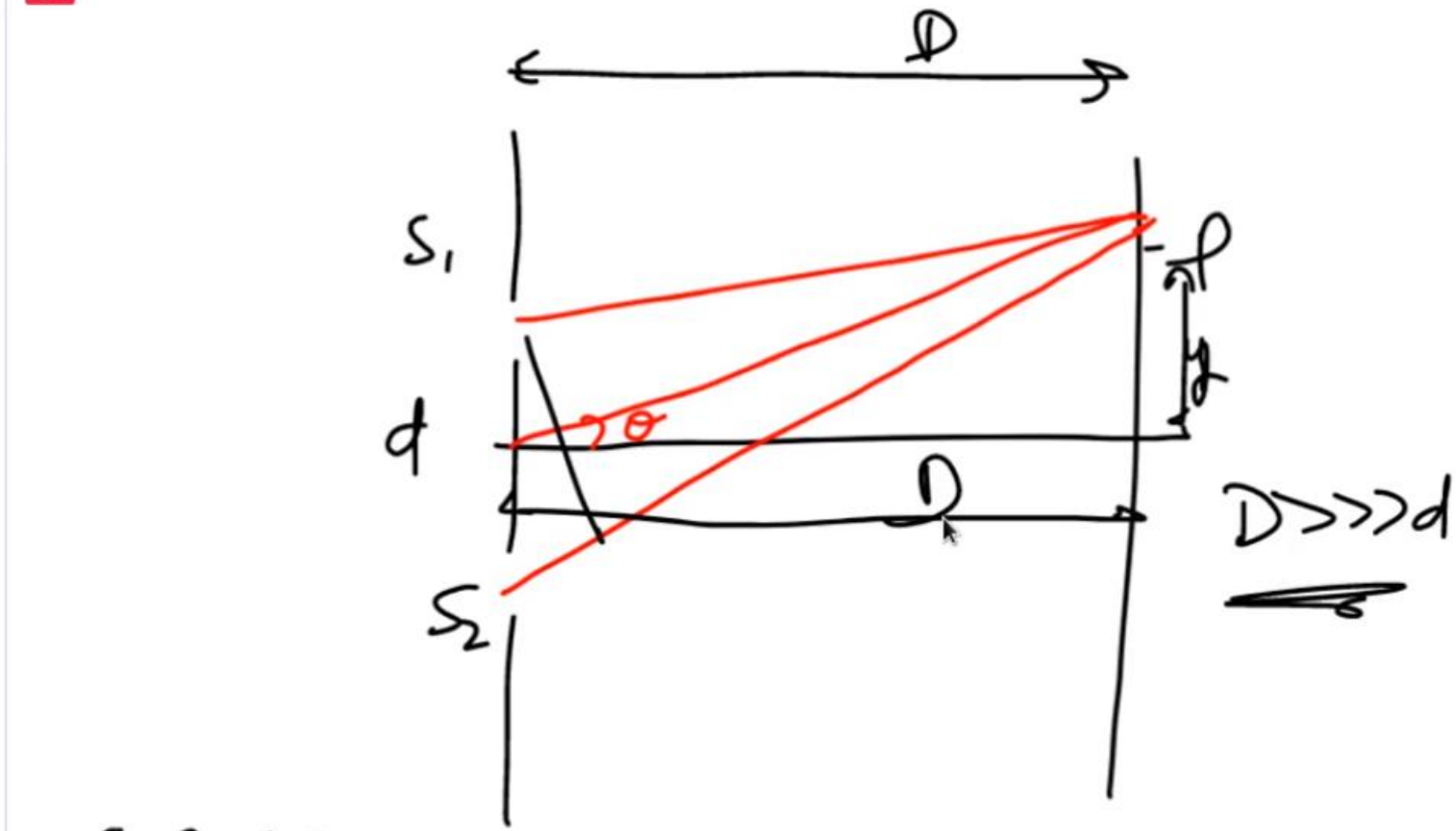
MA Madhav: Ohk

SA Sanyam kumar Jain: sir vaise techincally then we cant use sintheta = tantheta also

SR Srijahnavi Ch: here they took pi/6 = sin(pi/6)

Type a message

Chat Doubts More



$$S_2P - S_1P = d\sin\theta$$
$$= d\tan\theta = \frac{dy}{D}$$



techincally then
we cant use sintheta = tantheta also

SR Srijahnavi Ch
here they took pi/6 = sin(pi/6)

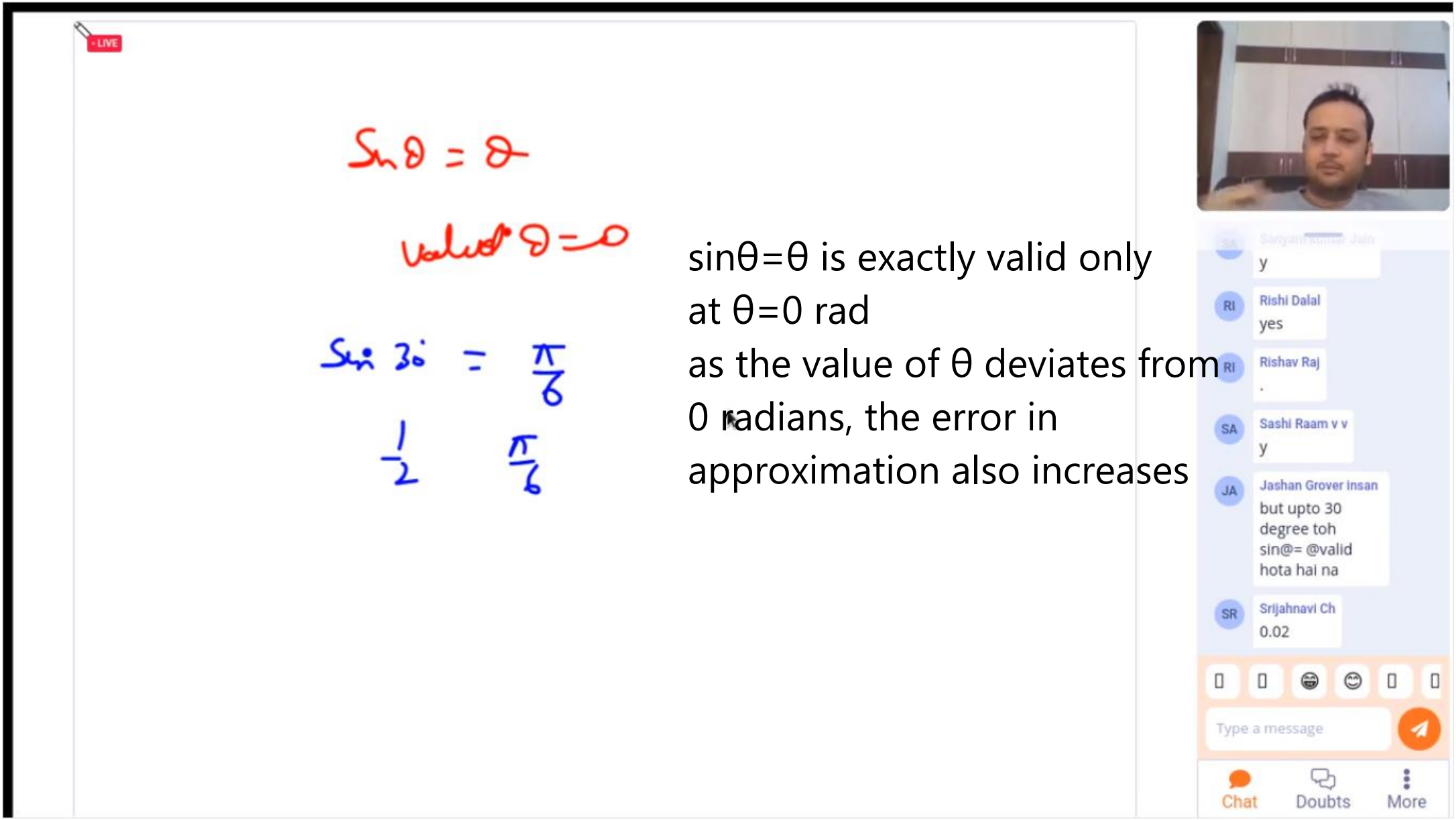
RI Rishav Raj
sir aur bsintheta =lemda approximation ni hota...?

SA Sanyam kumar Jain
ok sir clr

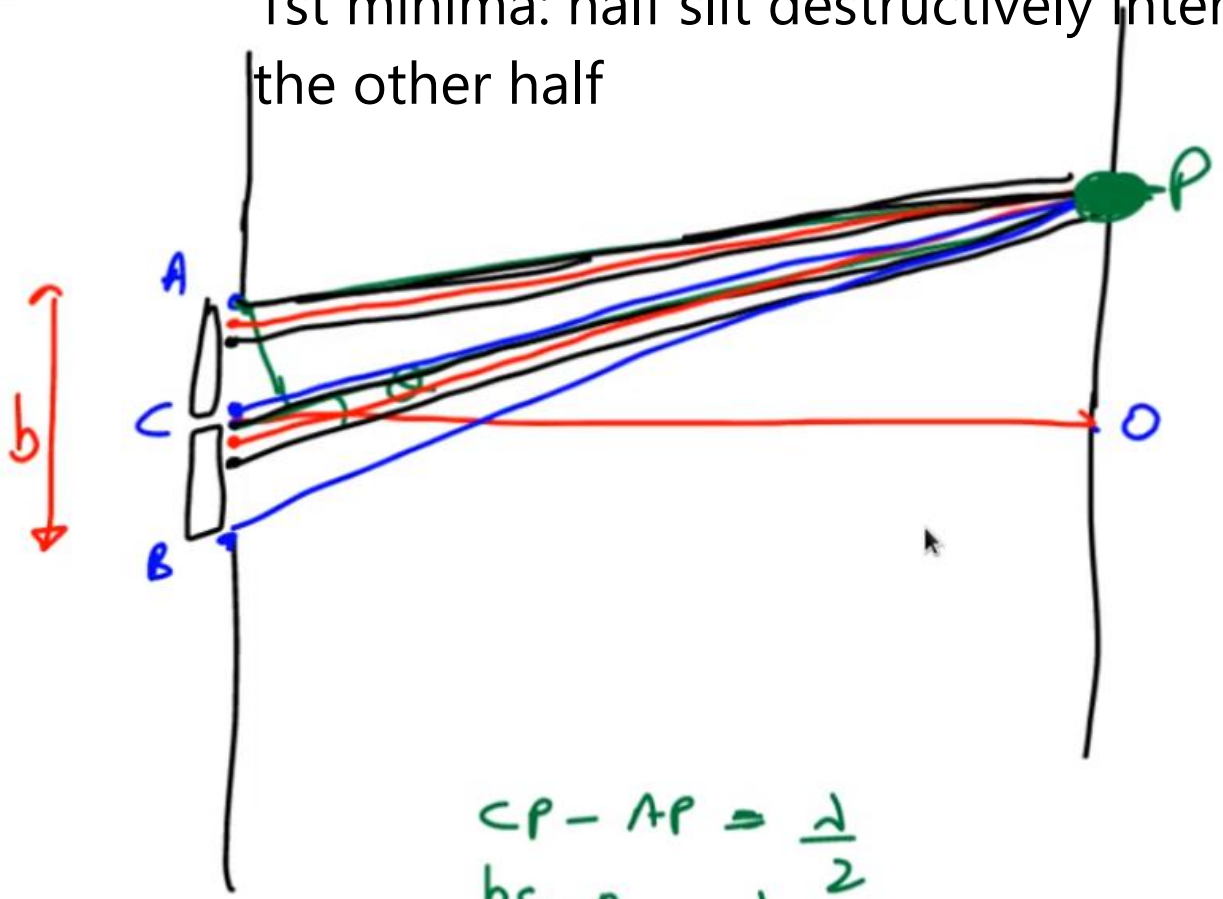
RI Rishav Raj
oo fir smjha

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Chat Doubts More


LIVE
sinθ = θ
valid θ = 0
sin 30° = π/6
1/2 π/6
sinθ=θ is exactly valid only at θ=0 rad
as the value of θ deviates from 0 radians, the error in approximation also increases
y
Rishi Dalal
yes
Rishav Raj
.
Sashi Raam v v
y
Jashan Grover Insan
but upto 30 degree toh sin@= @valid hota hai na
Srijahnavi Ch
0.02
Type a message
Chat
Doubts
More

1st minima: half slit destructively interferes with the other half



$$CP - AP = \frac{\lambda}{2}$$

$$\frac{b \sin\theta}{2} = \frac{\lambda}{2} \Rightarrow b \sin\theta = \lambda$$



y

SR Srijahnavi Ch
yes

JA Jashan Grover Insan
y

SA Sanyam kumar Jain
y

AB Abhiharsh Chauhan
ya

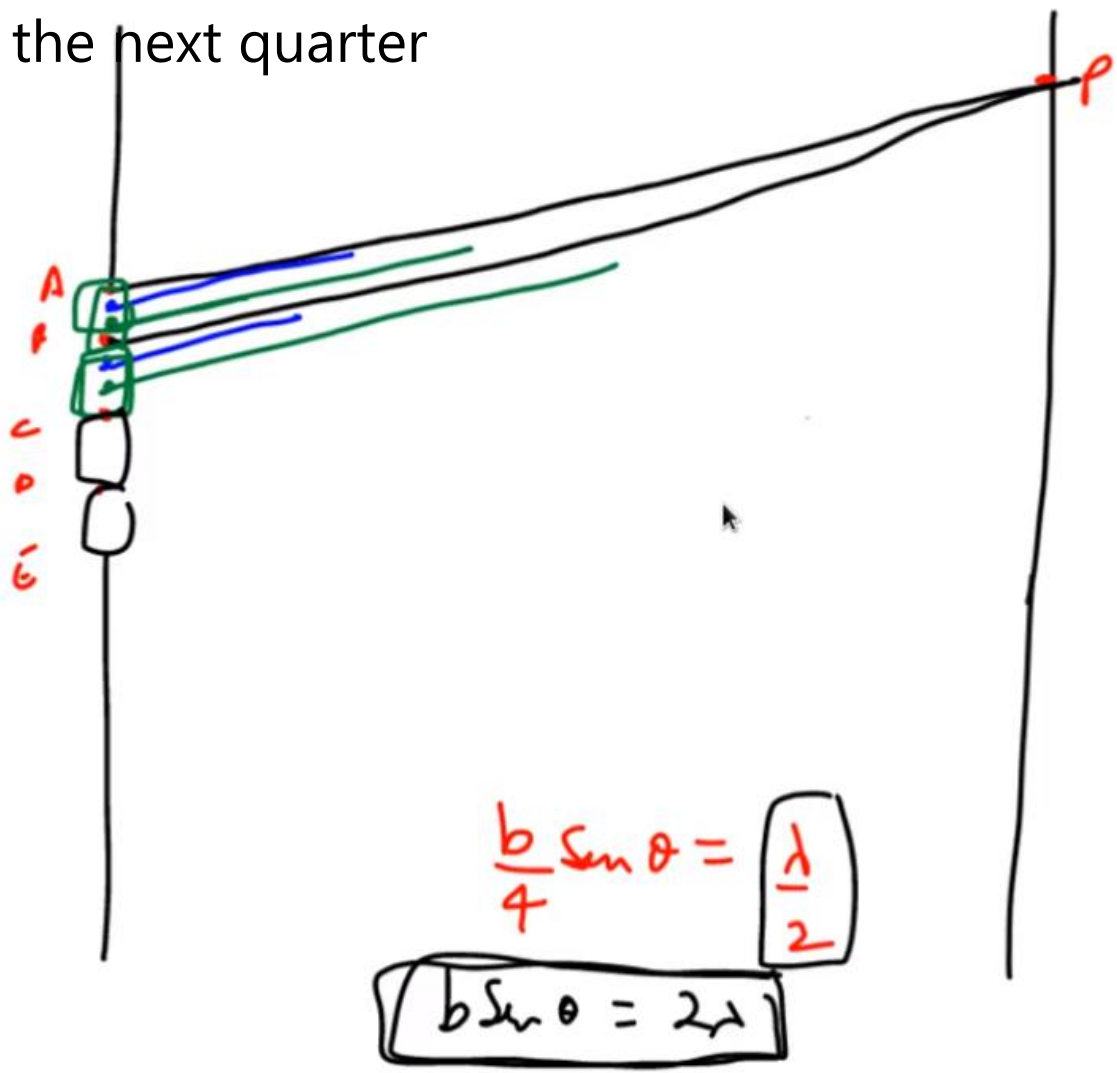
SA Sashi Raam v v
y

MA Manoj
Yes

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Chat Doubts More

2nd minima: quarter slit destructively interferes with the the next quarter



$$\frac{b}{4}\sin\theta = \frac{\lambda}{2}$$

$$b\sin\theta = 2\lambda$$



Anna Rathore
yes

RA Rachit Jain
Yes sir

SA Sanyam kumar Jain
sir toh aise ydse me single single slit se bhi ye rsult aana chiye na

JA Jashan Grover insan
y

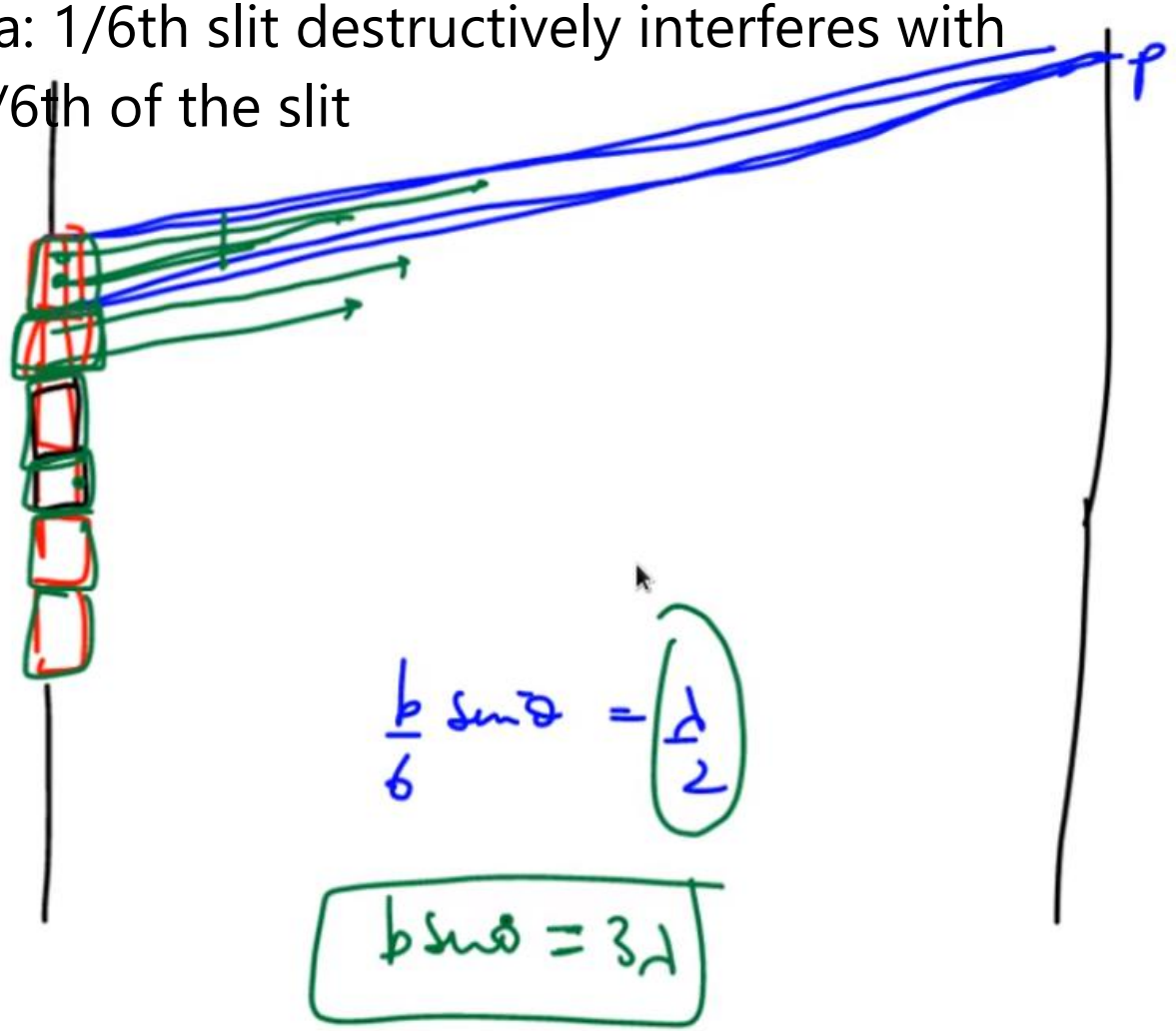
SA Sanyam kumar Jain
ok sir

SA Sanyam kumar Jain
clr sir

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Chat Doubts More

3rd minima: 1/6th slit destructively interferes with the next 1/6th of the slit





Sanyam kumar Jain
sir vaise aap flow me hindi bol rhe ho

Pranesh K.r
sir little more english

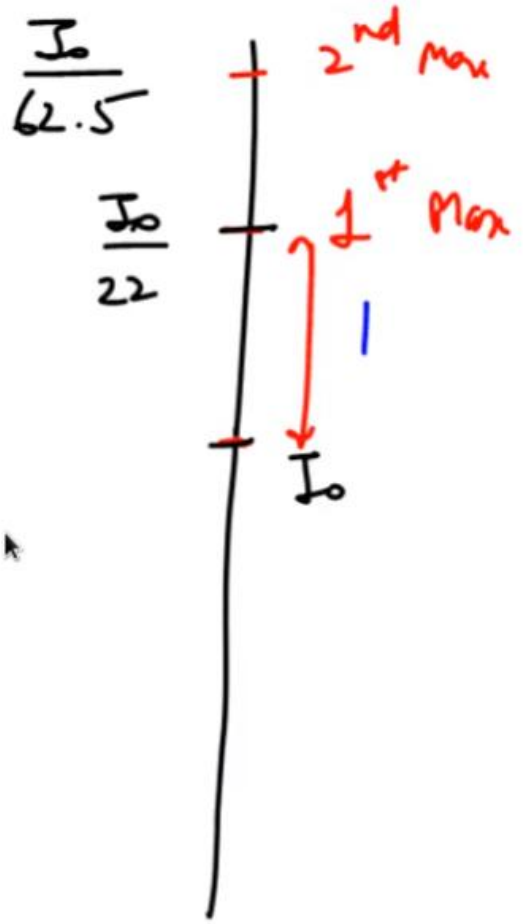
Sanyam kumar Jain

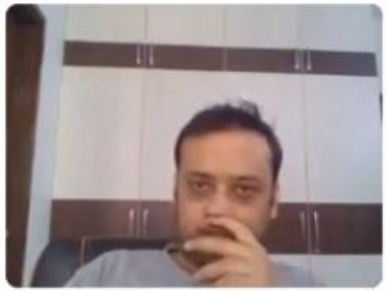
Sashi Raam v v
y

Jashan Grover insan
or we can also say b/2 Sin@= 3Lambda/2 ?

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beta

JA Jashan Grover Insan
2B

RI Rishi Dalal
DL/d

DE Deepak Chaurasia
0.5

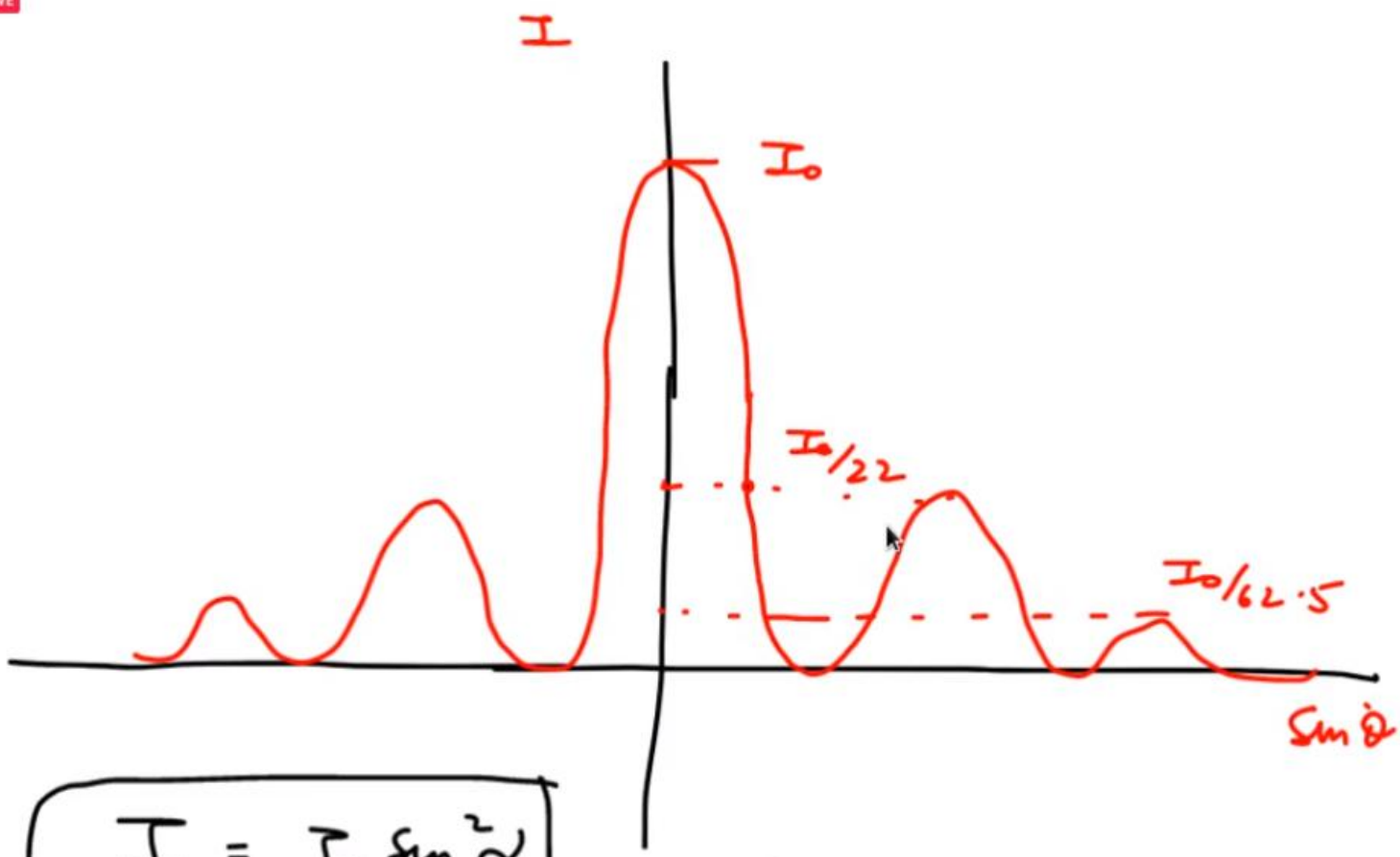
RI Rishav Raj
1.5

SA Sashi Raam v v
1.5B

RI Rishi Dalal
i thought min

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Chat Doubts More



$$I = I_0 \frac{\sin^2\alpha}{\alpha^2}$$

$$\alpha = \frac{\pi}{\lambda} b \sin\theta$$

for minima $\sin\theta = 0$ $\theta = n\pi$



Rishi Dalal
i thought min

Abhiharsh Chauhan
Sir Intensity ka func nahi yaad karke bas 3 maxima tak Intensity yaad kar le

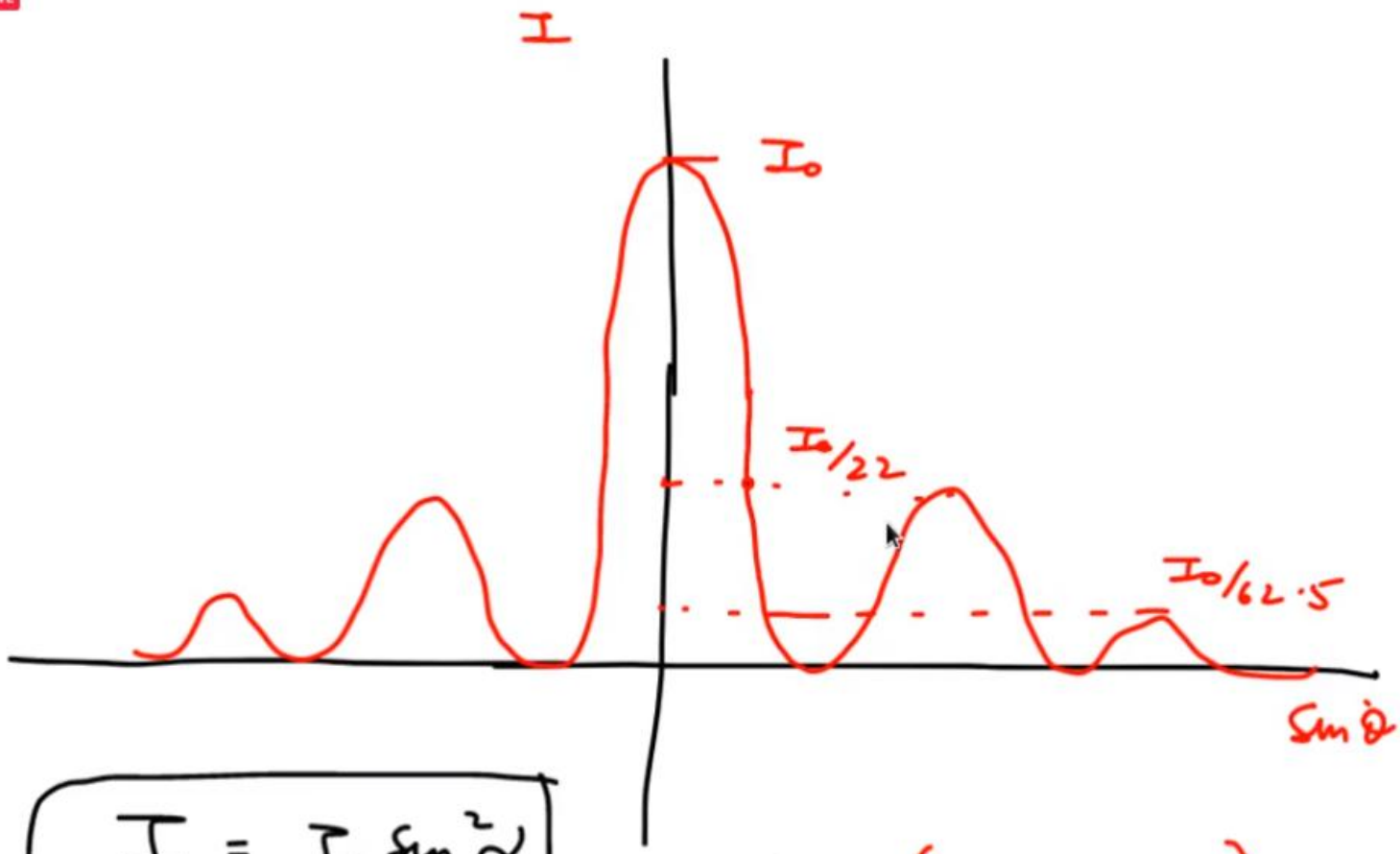
Rishi Dalal
psi

Abhiharsh Chauhan
ok Sir

Rishi Dalal
npi

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Chat Doubts More



$$I = I_0 \frac{\sin^2\alpha}{\alpha^2}$$

$$\alpha = \left(\frac{\pi}{\lambda} b \sin\theta\right) = n\pi$$

$$\alpha = n\pi \qquad b\sin\theta = n\lambda$$



Intensity yaad kar le

RI Rishi Dalal: psi

AB Abhiharsh Chauhan: ok Sir

RI Rishi Dalal: npi

SA Sashi Raam v v: yes

AB Abhiharsh Chauhan: ya

RI Rishi Dalal: yes

Type a message

Chat Doubts More

LIVE

$A_1 \sin(\omega t)$

$A_2 \sin(\omega t + \phi)$

$$A_{net} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos\phi}$$



yes

AB Abhiharsh Chauhan
ya

SA Sanyam kumar Jain
y

RI Rishi Dalal
facer

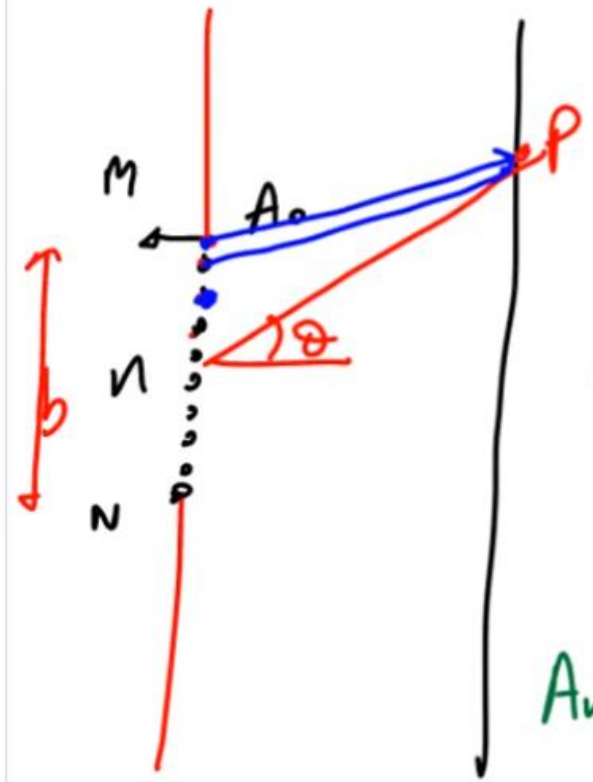
SA Sashi Raam v v
yes

JA Jashan Grover insan
y

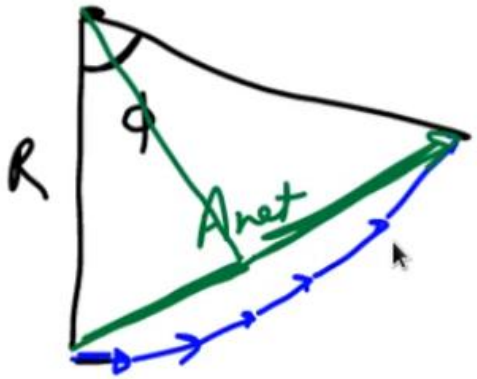
AM Amna Rahman
y

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$$\phi = b \sin\theta \frac{2\pi}{\lambda}$$



$$R\phi = nA_0$$

$$R\, b\sin\theta \frac{2\pi}{\lambda} = nA_0$$

$$A_{net} = 2R\sin\frac{\phi}{2}$$

$$= \frac{nA_0}{\frac{\pi}{\lambda} b\sin\theta} \sin\left(b\sin\theta \frac{\pi}{\lambda}\right)$$

$$= \frac{nA_0 \sin\alpha}{\alpha}$$



Abhiharsh Chauhan
ya

Amna Rahman
yes/

Sashi Raam v v
y

Rishav Raj
2Rsinphi/2

Rishi Dalal
ab square

Rishav Raj
haa I propnl to A^2 to aa jaaega

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Chat Doubts More

LIVE

$$I = k n^2 A_0^2 \frac{\sin^2 \alpha}{\alpha^2}$$

$$I = I_0 \frac{\sin^2 \alpha}{\alpha^2}$$



yes/

SA Sashi Raam v v
y

RI Rishav Raj
2Rsinphi/2

RI Rishi Dalal
ab square

RI Rishav Raj
haa I propnl to A^2 to aa jaaega

SA Sashi Raam v v
clr sir

DE Deepak Chaurasia
Y

Type a message

Chat Doubts More